GlandersWatch

Risk prediction, cited explanation and owner alerting for a neglected zoonosis

Pedapatruni Shanmuk Vardhan · Shaktimayee Mohanty · Yuvraj Singh

Supervisor: Dr. Sangita Dhara · IIIT Bhubaneswar

Overview

One probability, produced once, consumed in two different ways

1 Public data

Europe PMC · FAOSTAT
World Bank · NASA POWER

2 Risk engine

8 features → calibrated probability per country-year

3 Calibrated probability

A number that can be quoted: 0.03 means 3 in 100.

Explanation layer

Answers “why this country, and what should be done?” with passages retrieved from official guidance, cited, or refused.

Alert service

Raises a message only when the probability crosses a level, so a warning still means something on the day it arrives.

- Both consume the same calibrated output, so the message and the explanation can never disagree.

2,880
country-years, 2006–2023

160
countries

8
features

0.381
PR-AUC, unseen countries

Problem statement

Stated once, precisely, because every section inherits this framing

Given a country's features in year t, predict whether glanders is documented in that country in year t + 1.

The label

- One country-year is positive if a glanders occurrence is reported from that country in that year.
- Features come only from the year before, so the model never reads the outcome it is predicting.

The difficulty

- 169 positives in 2,880 rows. A model that says “never” is 94% accurate and useless.
- Accuracy and ROC-AUC hide this; precision-recall does not.

2,880	160	169	5.87%
country-years, 2006–2023	countries	positives	prevalence

Why it is needed

Surveillance tells you where the disease was, which is one step too late

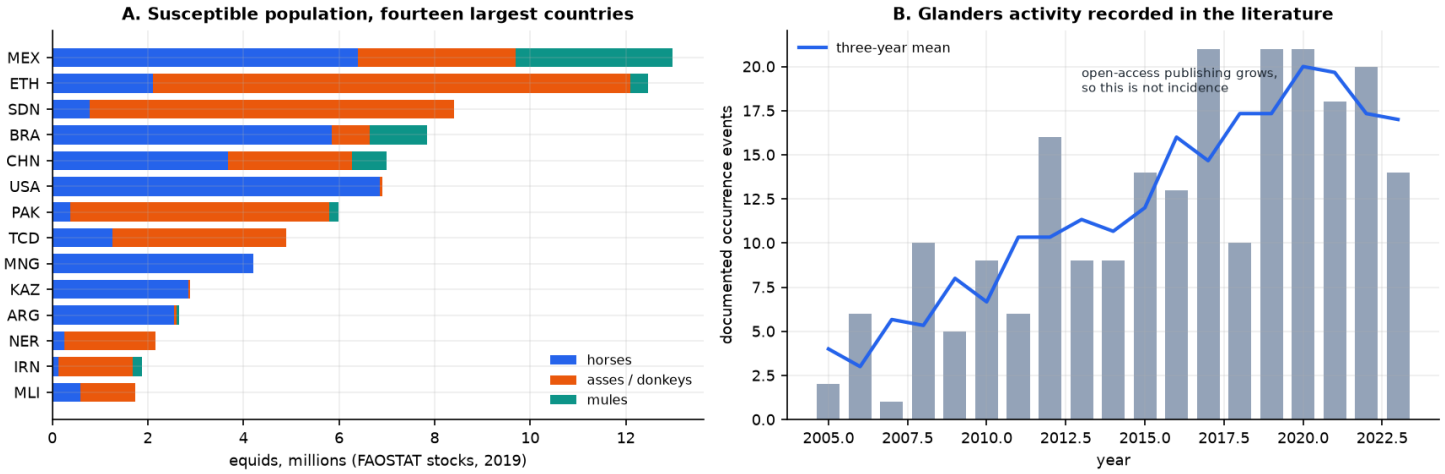
Who	What they cannot do today
District veterinary officer	Know where to pre-position testing and movement checks
Equine owner	Know whether this is the month to restrict movement
State planner	Compare risk across districts on one scale

What exists

- Reports that arrive after the fact. A national livestock forewarning system exists, but glanders is not on its disease list.

The failure mode to avoid

- A warning system that fires everywhere is ignored everywhere.



Panel A: how many equids are exposed. Panel B: how many events are documented per year. Documentation rises with open-access publishing, which is why the target is a documented event rather than raw incidence.

01

The risk engine

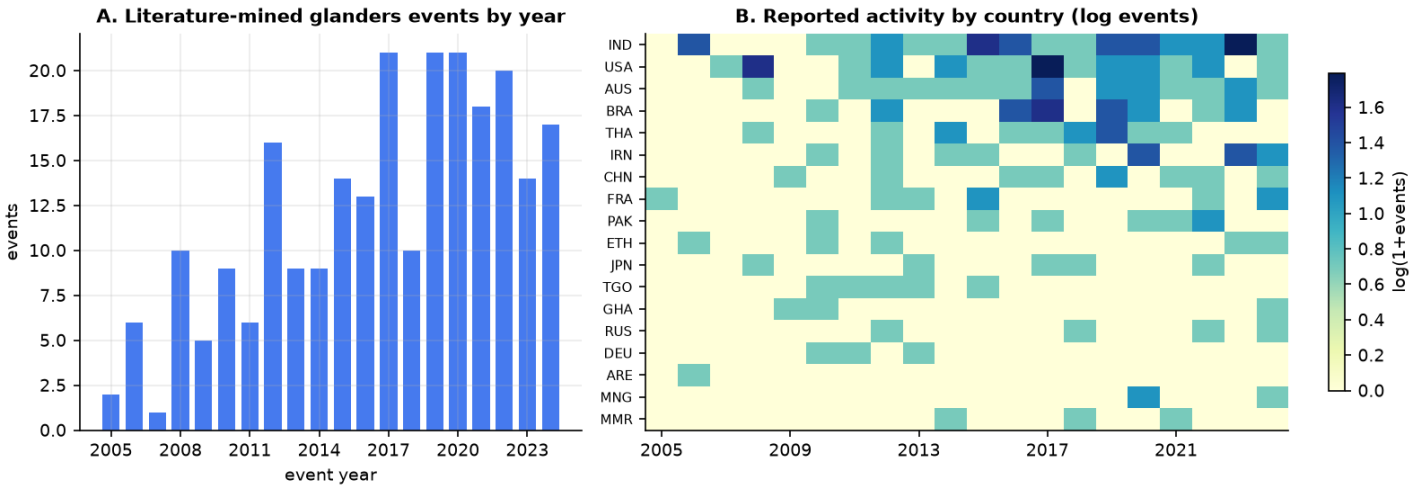
Where is glanders likely to be documented next year, and how much should we trust the number?

- Data: four public sources and the label they produce
- Eight features, chosen by attribution rather than association
- Two validation splits, and why the ranking inverts between them

Data

Four public sources, none previously applied to this disease

Source	What it contributes	Role
Europe PMC	Occurrence events from abstracts	The label
FAOSTAT	Horse, ass and mule head counts	Exposure
World Bank	Development and connectivity	Context
NASA POWER	Temperature, humidity, rainfall	Environment



Coverage of the assembled panel: which countries contribute which years, and where the positives sit.

- 6,700 raw country-year rows over 268 reporting entities.
- Filtered to 2,880 rows across 160 equine-holding countries.
- 169 positives, 5.87% prevalence.
- Rows without a climate baseline are dropped, not imputed, so no anomaly is manufactured.

Eight features, chosen by attribution

Ranked by mean absolute SHAP on the training years only

#	Feature	Unit	Mean |SHAP|
1	pop_total	persons	1.035
2	years_since_last	years (99 = none)	0.648
3	temp_mean_anom	°C from baseline	0.505
4	internet_pct	% of population	0.449
5	humidity_anom	% RH from baseline	0.382
6	precip_cv	ratio, sd/mean	0.371
7	log_livestock	ln(1 + LSU)	0.364
8	goats	head	0.362

- 8 of 42 candidates, carrying 49% of total attribution.
- Selection uses training years only, so held-out years cannot shape the choice.
- Correlation puts longitude first; attribution puts it near the bottom.

Attribute a prediction

SHAP says how much each feature moved this model's output, so the ranking matches what was actually used.

Beyond pairs

Correlation and mutual information score features pairwise; they cannot see interactions or redundancy between features.

Comparable

Both are unnormalised and undirected. Attribution is additive and sums to the prediction.

Results

One target, two validation splits, opposite conclusions

Model	PR-AUC, unseen years	PR-AUC, unseen countries
Logistic Regression	0.461	0.381
Decision Tree	0.376	0.243
Random Forest	0.495	0.322
Extra Trees	0.494	0.353
XGBoost	0.429	0.268
LightGBM	0.471	0.280

- Base rate 5.87%, so PR-AUC is the metric that moves.
- ROC-AUC stays in 0.80–0.87 for all six models on both splits; it cannot separate them.
- Logistic regression is the champion: it wins on unseen countries, which is what deployment means.

0.495
random forest, unseen years

0.381
logistic, unseen countries

0.322
random forest, unseen countries

The ranking inverts. Random forest wins on unseen years; logistic regression wins on unseen countries — and deployment means an unseen country, so the simpler model is the one shipped.

The number is trustworthy, and that sets the alert budget

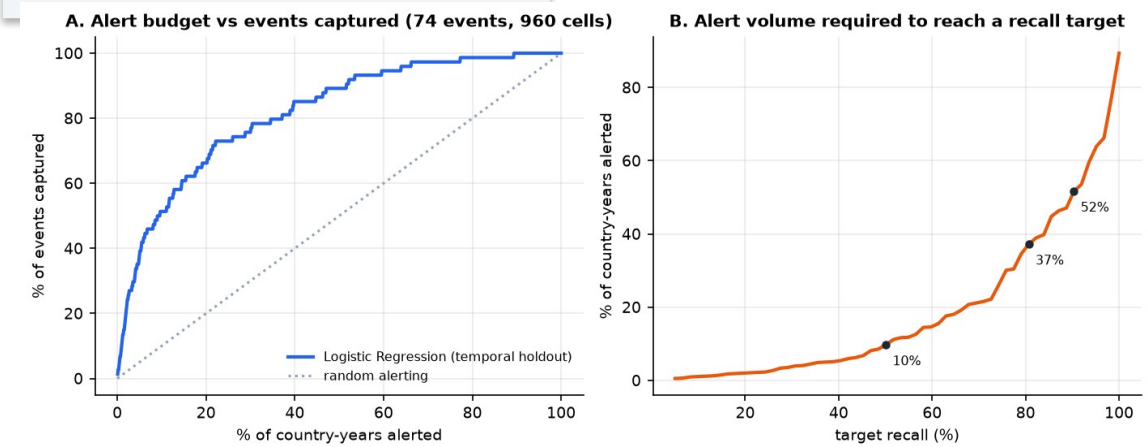
Calibration is what lets a message quote a probability at all

0.025
expected calibration error

0.056
Brier score

43%
events caught, alerting 6% of country-years

- Half of all events are reached by alerting 9% of country-years; 80% by 37%.
- That curve, not a default 0.5 cutoff, is where alert levels come from.
- It is also the argument that the alert service can be quiet without being useless.



This curve is the link between the two halves of the system: it states how quiet an alert service can be while still catching most events.

02

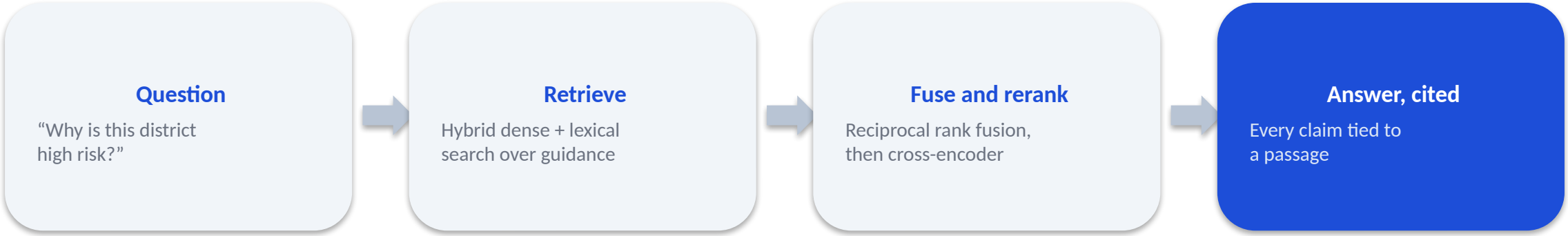
The explanation layer

Why did the model flag this country, and what should an owner actually do about it?

— Grounded retrieval with citations, and refusal when nothing supports an answer
— The guardrail that keeps advisory text from inventing guidance
— How this layer will be measured, and what is not yet measured

How an answer is built

Retrieval first, generation second, citation always



Guardrails, not suggestions

- Refuses to answer outside the retrieved sources, and refuses to diagnose an individual animal.
- Always shows the passages it used.

Why retrieval and not a plain model

- Guidance changes; weights would have to be retrained to follow it.
- A cited passage can be checked by a veterinarian. A generated sentence cannot.

“Risk is elevated mainly because of recent reported activity in neighbouring districts and high equine density. Restrict equid movement and contact the local veterinary authority if an animal is suspected.” — each clause carries its source; where no passage supports the question, the layer says so rather than answering.

How this layer will be judged

The metrics are defined now; measuring them is the next phase

Metric	What it asks	How
Recall@k	Is the right passage in the top k?	Hand-built question set
Faithfulness	Does the answer follow from those passages?	RAGAS, plus manual review
Answer relevance	Does it answer the question asked?	RAGAS
Citation precision	Does every claim trace to a passage?	Automated check

- These four cover the ways an explanation fails: retrieving the wrong passage, drifting from the passages it retrieved, or faithfully answering a question nobody asked.

None of these are measured yet. Targets will be fixed before measurement, so the layer cannot be scored by choosing a threshold after seeing the results.

03

The alert service

How does a probability become a message that is still read in month twelve?

- Levels taken from the calibration curve, not a default cutoff
- Hysteresis, so an area does not flicker between levels
- Three layers of deduplication, and a stale-data guard
- How delivery will be measured, and what is not yet measured

How an alert is raised

Five stages; each one exists to stop a specific failure

1 Predict
Latest calibrated score for the area

2 Assign level
Thresholds taken from the budget curve

3 Deduplicate
Three idempotency keys, not one

4 Deliver
SMS or WhatsApp, with retries

5 Log
Status per recipient, for the admin

Hysteresis

Enter at a threshold, leave only 0.05 below it. A country sitting on the line does not oscillate.

Freshness

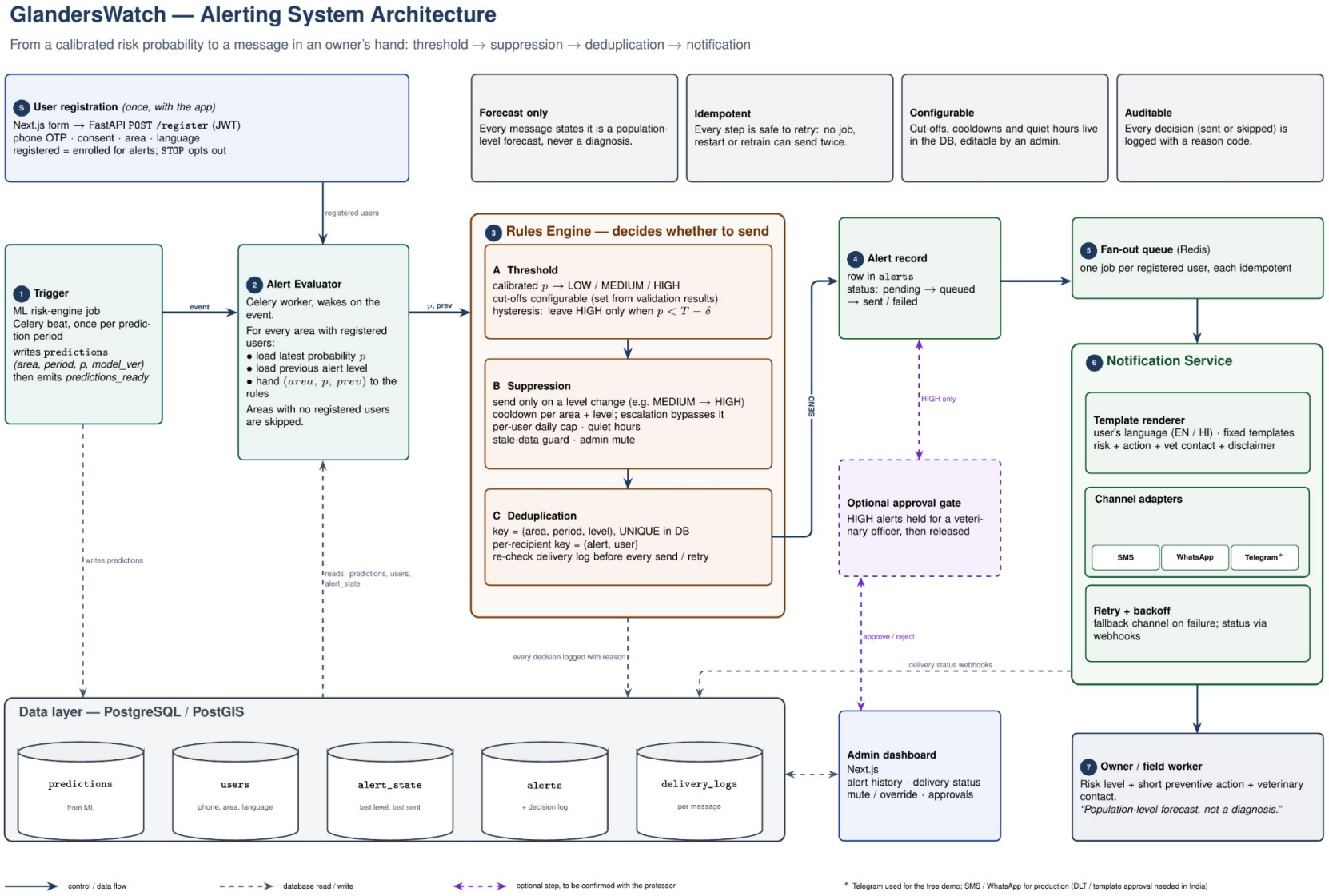
If the data check fails, no alert goes out and the admin is told instead.

Quiet hours

Hold overnight except for escalation, so a real rise still gets through.

Levels come from the calibration curve, so “high” has a measured meaning rather than a default cutoff of 0.5. Deduplication is what keeps a re-run or a retrain from sending the same warning twice.

The alert pipeline, end to end



The deployed view: evaluator, levels, deduplication, delivery and logging, with an admin surface for failures.

How the alert service will be judged

Because the cost of a false alarm is measured in attention, not accuracy

Metric	What it asks
Precision	Of the alerts sent, how many were real events?
Recall	Of the real events, how many did we alert on?
Missed events	The failures a user actually feels
False alerts per subscriber	The alert-fatigue budget

- Defined but not yet measured.
- A pilot with a district office is how they get numbers.

GLANDERS RISK ALERT — Risk: HIGH. Elevated glanders risk is predicted for your registered area for the next period. Reason: recent reported activity and environmental factors. Action: avoid unnecessary movement or mixing of equids; contact the local veterinary authority if an animal is suspected. This is a population-level forecast, not a diagnosis.

Where the project stands

What is built, what is specified, and what a pilot would settle

Built and measured

Risk engine. Six models on two splits. Champion is logistic regression, PR-AUC 0.381 on unseen countries, ECE 0.025.

Specified, not deployed

Explanation layer with citation checks and refusal. Alert service with levels, hysteresis and deduplication. Both have metrics defined, neither has numbers yet.

What a pilot would settle

Alert precision and fatigue against a real district office. Explanation faithfulness on a hand-built question set. A label audit on a stratified sample.

The model is the part that is finished. The evidence that the explanation and alert layers behave in use is the part that is not, and a pilot is how it arrives.

- Known limits throughout: the label is documented activity, not incidence, and with 169 positives the uncertainty on every headline metric is wide.